

Steps:

1. Downloaded pre-event COMET-LiCSAR interferograms (frame 062D_05831_131313, descending) with download_licsar_pre_event.py - cutoff 20170624, 72-day max baseline.
2. Subset to AOI (103.62-103.70E, 32.05-32.11N) and converted to a MintPy stack with licsar_to_mintpy_h5.py.
3. Ran SBAS via smallbaselineApp.py.
4. Chose the reference point: started with auto - got a scene-wide seasonal artifact - moved to a stable, high-coherence pixel just outside the slide (a low-coherence "stable" pixel had degraded the whole solution).
5. Extracted the velocity map and a landslide-pixel time series.
6. Opened the coherence (.cc) file for the event-spanning 20170607–20170725 interferogram.
7. Computed Sentinel-1 VV change (post - pre, dB) in Google Earth Engine.
8. Built before/after S2 true-colour composites to map the scar.
9. Computed an NDVI difference to quantify vegetation change.

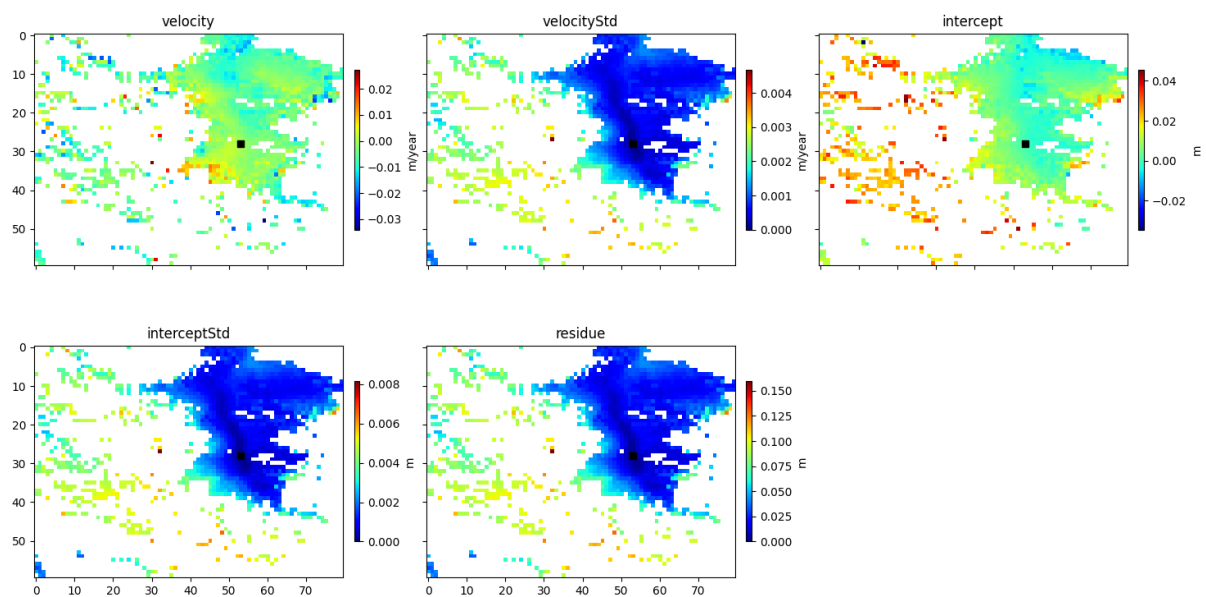


Figure 1 SBAS results over AOI

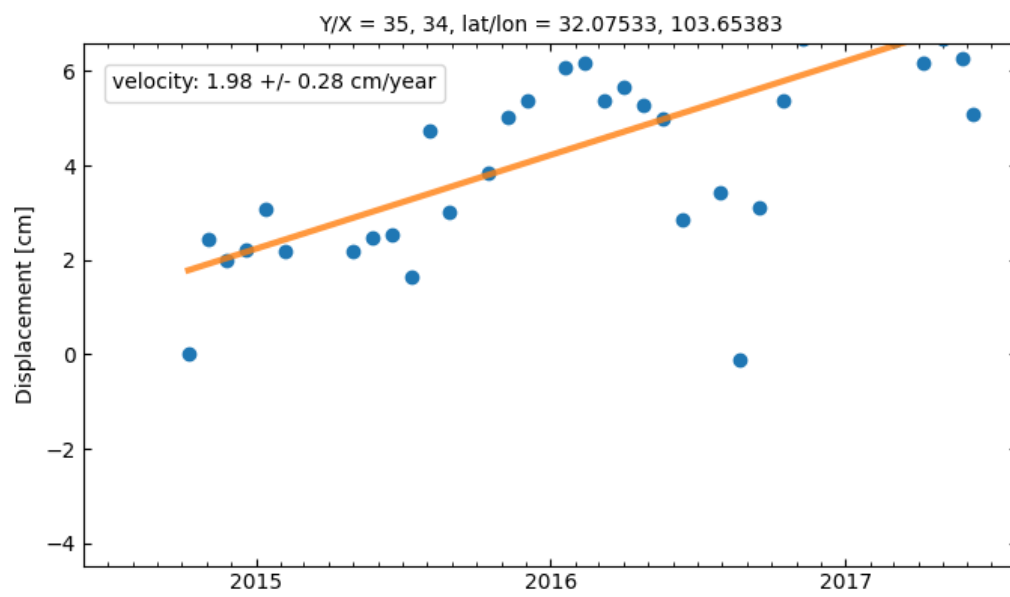


Figure 2 LOS displacement timeseries for landslide pixel

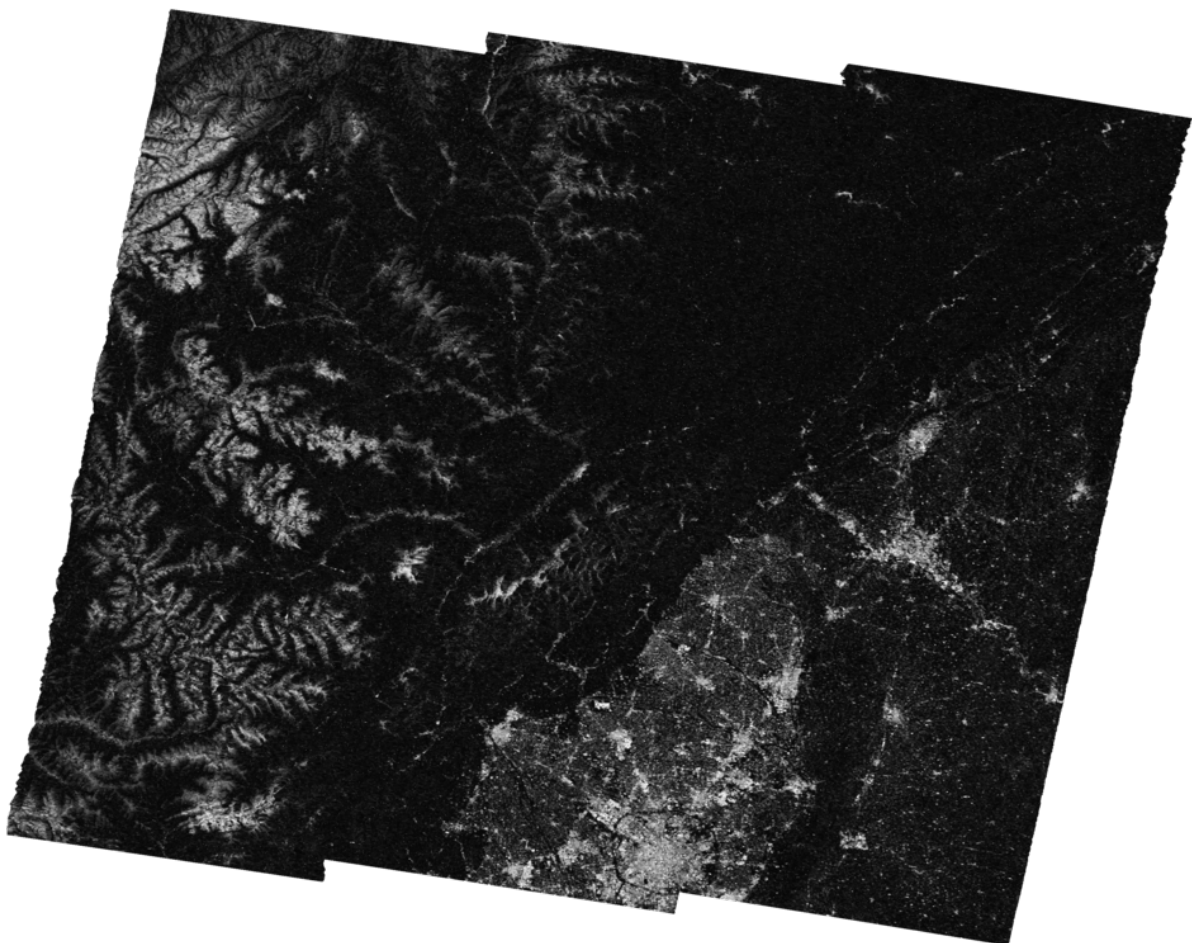


Figure 3 Interferometric coherence for the event

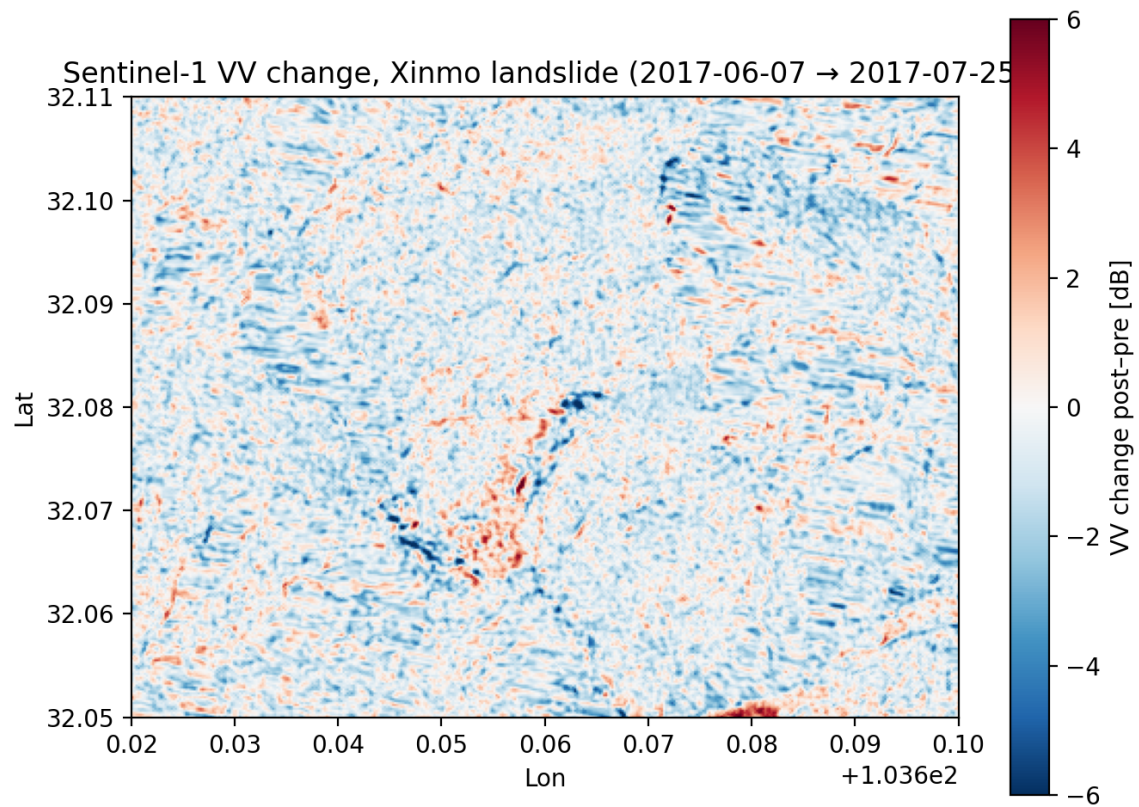


Figure 4 Sentiner-1 VV backscatter change (post – pre)

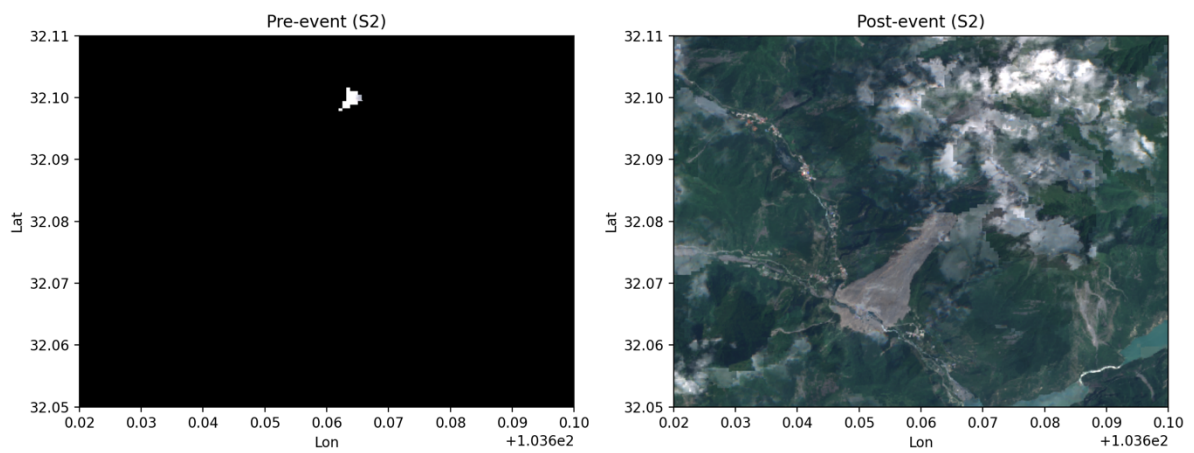


Figure 5 Sentinel-2 true colour composites before (left) and after (right)

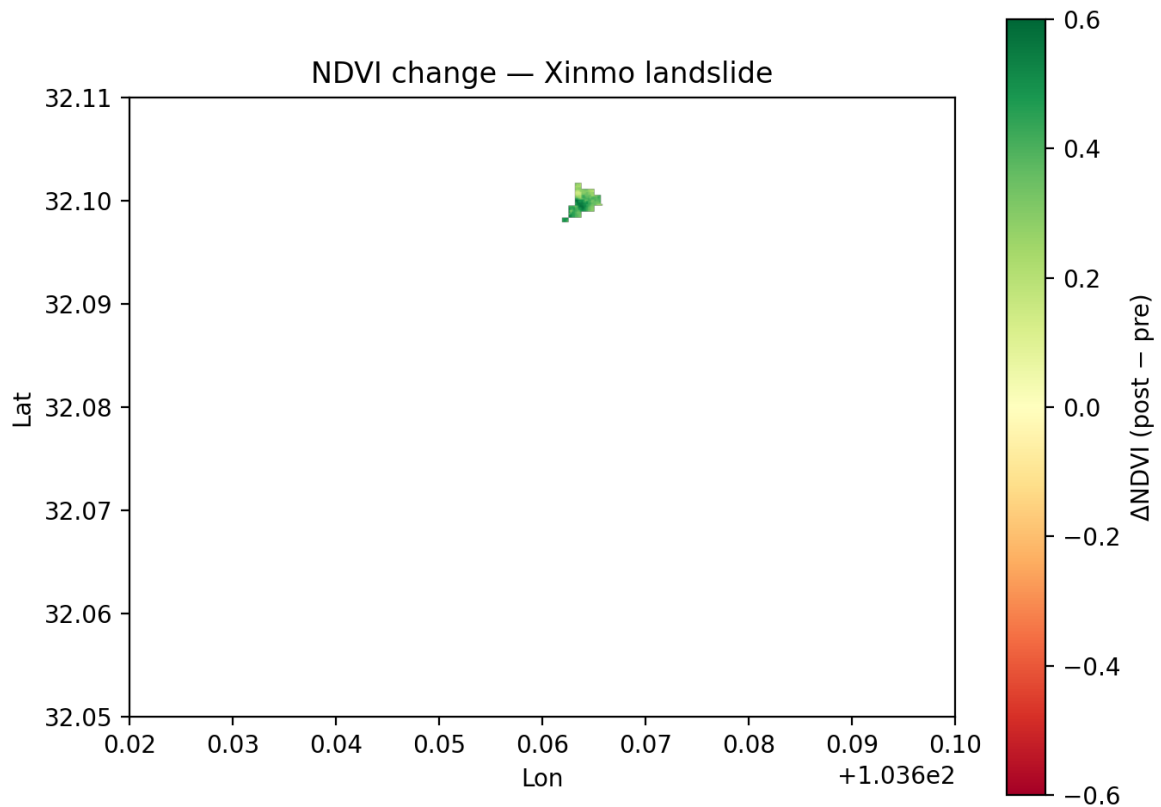


Figure 6 NDVI change (post - pre)

Summary

InSAR time series: the slope showed slow precursory creep of $\sim 1.98 \pm 0.28$ cm/yr in LOS. Coherence – not useful in this case, only city/ valley stays coherent. VV change – better than coherence, but heavily contaminated by speckle and topography. Optical – the post-event S2 image maps the failure cleanly. NDVI – missing dry season (heavy cloud contamination – monsoon season) baseline. Overall each method has its problems, so the best way is to combine all of them to obtain better results.